

FIGURE P3-9

- 3-10 (a) Formulate mesh-current equations for the circuit in Figure P3-10.
 (b) Use these equations to find v_x and i_x .

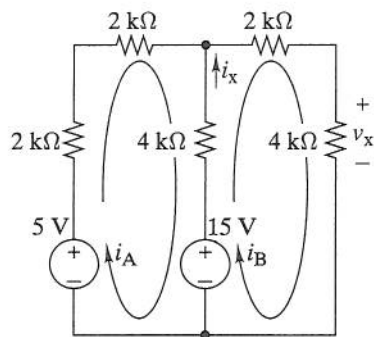


FIGURE P3-10

- 3-11 (a) Formulate mesh-current equations for the circuit in Figure P3-11.
 (b) Use these equations to find v_x and i_x .

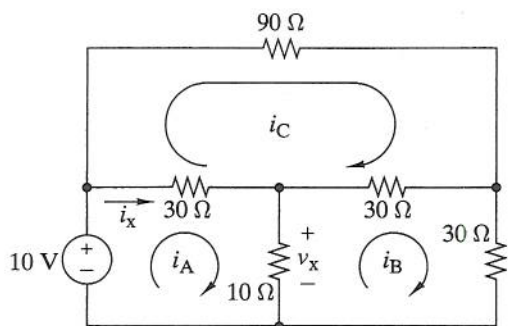


FIGURE P3-11

- 3-12 (a) Formulate mesh-current equations for the circuit in Figure P3-12.
 (b) Solve for v_x and i_x when $R_1 = 200 \Omega$, $R_2 = 300 \Omega$, $R_3 = 50 \Omega$, $R_4 = 250 \Omega$, $R_5 = 200 \Omega$, $i_s = 50 \text{ mA}$, and $v_s = 15 \text{ V}$.
 (c) Find the total power dissipated in the circuit.

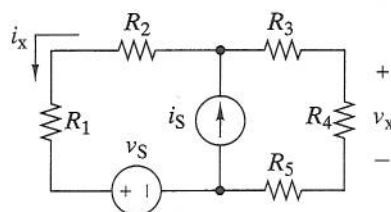


FIGURE P3-12

- 3-13 (a) Formulate mesh-current equations for the circuit in Figure P3-13.
 (b) Solve for v_x and i_x when $R_1 = R_2 = 10 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$, $R_4 = 1 \text{ k}\Omega$, $i_s = 2.5 \text{ mA}$, $v_{s1} = 12 \text{ V}$, and $v_{s2} = 0.5 \text{ V}$.
 (c) Find the power supplied by v_{s1} .

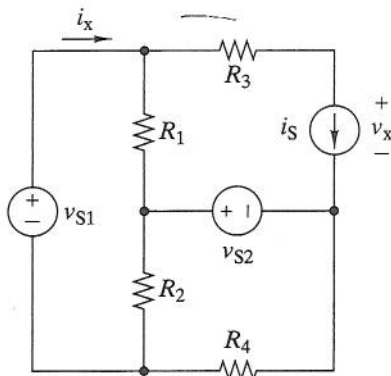


FIGURE P3-13

- 3-14 The circuit in Figure P3-14 seems to require two supermeshes since both current sources appear in two meshes. However, sometimes rearranging the circuit diagram will eliminate the need for a supermesh.
 (a) Show that supermeshes can be avoided in Figure P3-14 by rearranging the connection of resistor R_6 .
 (b) Formulate mesh-current equations for the modified circuit as redrawn in part (a).
 (c) Solve for v_x when $R_1 = R_2 = R_3 = R_4 = 2 \text{ k}\Omega$, $R_5 = R_6 = 1 \text{ k}\Omega$, $i_{s1} = 40 \text{ mA}$, and $i_{s2} = 20 \text{ mA}$.

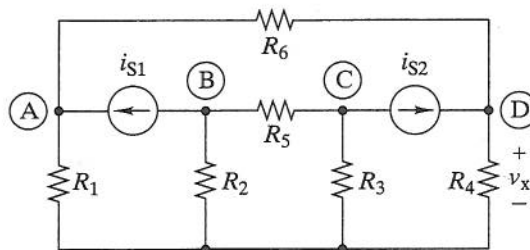


FIGURE P3-14

- 3-15 (a) Formulate mesh-current equations for the circuit in Figure P3-15.
 (b) Use these equations to find v_x and i_x .
 (c) Find the total power dissipated in the circuit.



FIGURE P3-15

- 3-16 (a) Formulate mesh-current equations for the circuit in Figure P3-16.
 (b) Use these equations to find v_x and i_x .

FIGURE P3-16

- 3-17 In Figure P3-17, find the voltage at node C relative to the reference node and the mesh current i_x .

FIGURE P3-17

- 3-18 In Figure P3-18, find the voltage at node A relative to the reference node and the mesh current i_x .

FIGURE P3-18

- 3-15 (a) Formulate mesh-current equations for the circuit in Figure P3-15.
 (b) Use these equations to find v_x and i_x .
 (c) Find the total power delivered to the resistors.

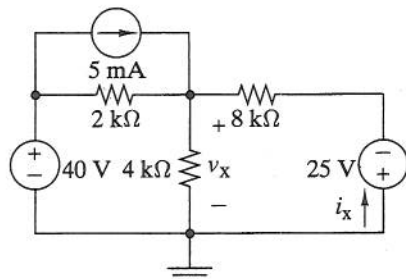


FIGURE P3-15

- 3-16 (a) Formulate mesh-current equations for the circuit in Figure P3-16.
 (b) Use these equations to find the input resistance.

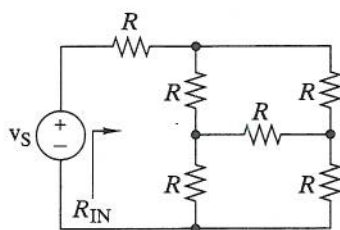


FIGURE P3-16

- 3-17 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

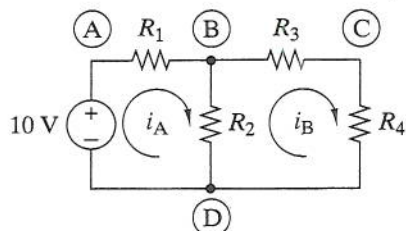


FIGURE P3-17

- 3-18 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$. The voltage at node A is observed to be $v_A = 8\text{ V}$ when node C is connected to ground. Find the node voltages v_B and v_D , and the mesh currents i_A and i_B .

- 3-19 Find the node voltages v_A and v_B in Figure P3-19.

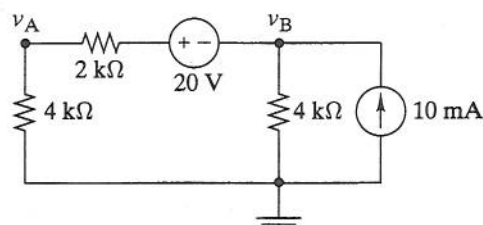


FIGURE P3-19

- 3-20 Find the mesh currents i_A , i_B , and i_C in Figure P3-20.

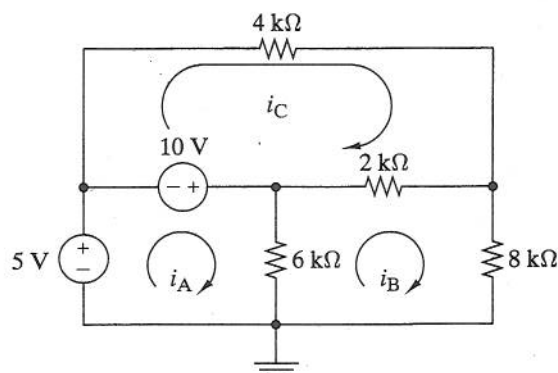


FIGURE P3-20

OBJECTIVE 3-2 LINEARITY PROPERTIES (SECT. 3-3)

- (a) Given a circuit containing linear resistors and one independent source, use the proportionality principle to find selected signal variables.
 (b) Given a circuit containing linear resistors and two or more independent sources, use the superposition principle to find selected signal variables.
 See Examples 3-10, 3-11, 3-12 and Exercises 3-11, 3-12, 3-13, 3-14, 3-15

- 3-21 Find the proportionality constant $K = v_O/i_S$ for the circuit in Figure P3-21.

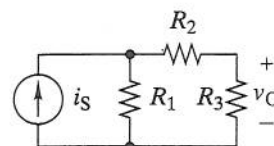


FIGURE P3-21

- 3-22 Find the proportionality constant $K = i_O/v_S$ for the circuit in Figure P3-22.

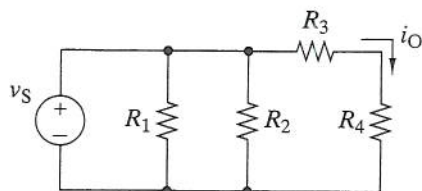


FIGURE P3-22

- 3-23 Find the proportionality constant $K = v_O/i_S$ for the circuit in Figure P3-23.

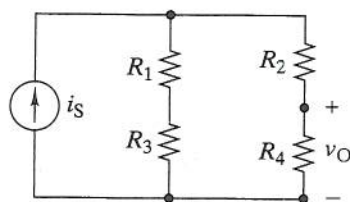


FIGURE P3-23

- 3-24 Use the unit output method to find v_O in Figure P3-24.

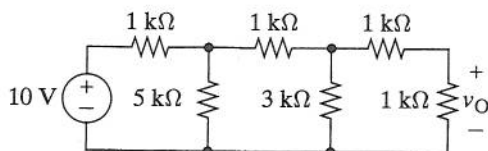


FIGURE P3-24

- 3-25 Use the unit output method to find i_O in Figure P3-25.

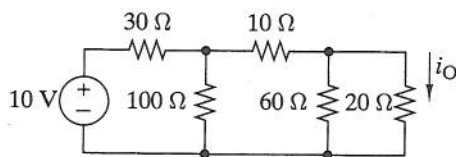


FIGURE P3-25

- 3-26 Use the superposition principle to find v_O in Figure P3-26.

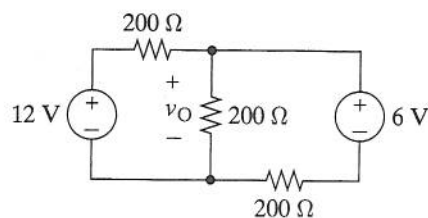


FIGURE P3-26

- 3-27 Use the superposition principle to find i_O in Figure P3-27.

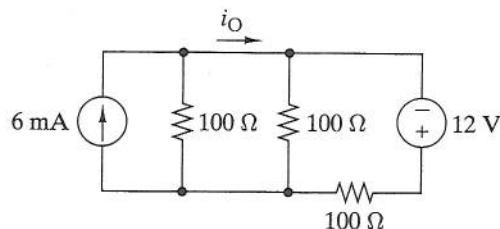


FIGURE P3-27

- 3-28 Use the superposition principle to find v_O in Figure P3-28.

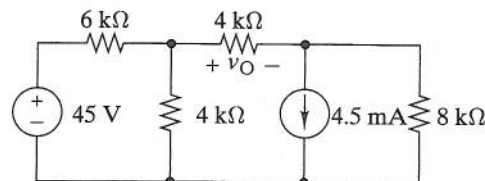


FIGURE P3-28

- 3-29 Use the superposition principle to find i_O in Figure P3-29.

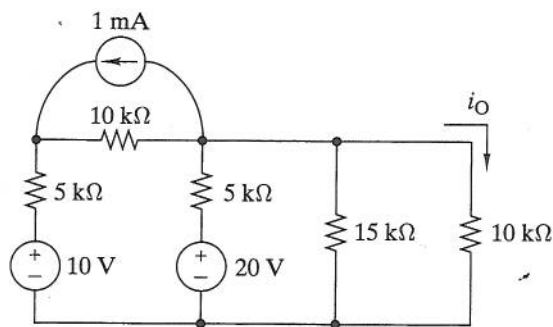


FIGURE P3-29

3-30 Use the superposition principle to find v_O in terms of v_1 , v_2 , and R in Figure P3-30.

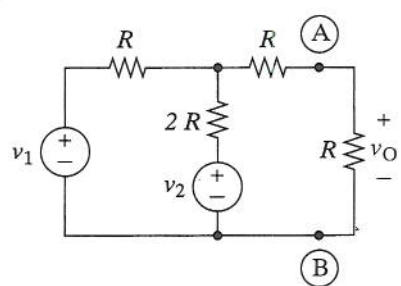


FIGURE P3-30

3-31 Use the superposition principle to find v_O in terms of i_1 , i_2 , and R in Figure P3-31.

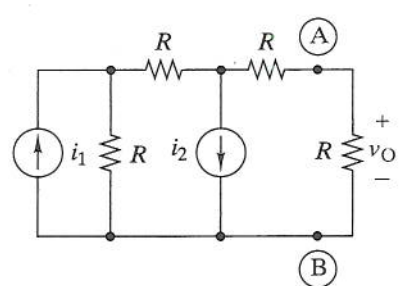


FIGURE P3-31

3-32 A linear circuit containing two sources drives a $100\text{-}\Omega$ load resistor. Source number 1 delivers 250 mW to the load when source number 2 is off. Source number 2 delivers 4 W to the load when source number 1 is off. Find the power delivered to the load when both sources are on. *Hint:* The answer is not 4.25 W . Why?

3-33 A linear circuit is driven by an independent voltage source $v_S = 10\text{ V}$ and an independent current source $i_S = 10\text{ mA}$. The output voltage is $v_O = 2\text{ V}$ when the voltage source is on and the current source off. The output is $v_O = 1\text{ V}$ when both sources are on. Find the output voltage when $v_S = 5\text{ V}$ and $i_S = -10\text{ mA}$.

3-34 A certain linear circuit has three input voltages and one output voltage v_O . The following table lists the output for different values of the three inputs. Find the input-output relationship for the circuit.

$v_{S1}(\text{V})$	$v_{S2}(\text{V})$	$v_{S3}(\text{V})$	$v_O(\text{V})$
2	4	-4	1
1	2	2	1.5
1	4	2	2

3-35 This problem involves designing a resistive circuit with two inputs v_{S1} and v_{S2} and a single output voltage v_O . Design the circuit so that $v_O = K(v_{S1} + 4v_{S2})$ is delivered across a $500\text{-}\Omega$ load. The value of K is not specified but should be greater than $1/10$.

OBJECTIVE 3-3 THÉVENIN AND NORTON EQUIVALENT CIRCUITS (SECT. 3-4)

Given a circuit containing linear resistors and independent sources:

- (a) Find the Thévenin or Norton equivalent at a specified pair of terminals.
 - (b) Use the Thévenin or Norton equivalent to find the signals delivered to linear or nonlinear loads.
- See Examples 3-13, 3-14, 3-15, 3-16 and Exercises 3-16, 3-17, 3-18

3-36 (a) Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-36.

(b) Use the equivalent circuit found in part (a) to find i_L in terms of i_S , R_1 , R_2 , and R_L .

(c) Check your answer in part (b) using current division.

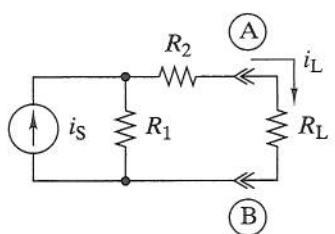


FIGURE P3-36

3-37 Find the Thévenin equivalent circuit seen by R_L in Figure P3-37. Find the voltage across the load when $R_L = 5\text{ }\Omega$, $10\text{ }\Omega$, and $50\text{ }\Omega$.

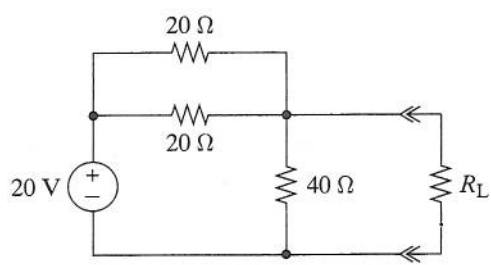


FIGURE P3-37

3-38 Find the Norton equivalent seen by R_L in Figure P3-38. Find the current through the load when $R_L = 6\text{ }\Omega$, $12\text{ }\Omega$, and $60\text{ }\Omega$.

analysis or design of complex OP AMP circuits can be based on these four building blocks provided the interconnections are made between the output of one to the input of another.

- Important applications of OP AMPs include digital-to-digital converters, transducer interface circuits, and comparator circuits.

PROBLEMS

OBJECTIVE 4-1 LINEAR ACTIVE CIRCUITS (SECTS. 4-1, 4-2)

Given a circuit containing linear resistors, dependent sources, and independent sources, find selected output signal variables, input-output relationships, or input-output resistances. See Examples 4-2, 4-3, 4-4, 4-5, 4-6, 4-7, 4-8 and Exercises 4-1, 4-2, 4-3, 4-5, 4-6

- 4-1 Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4-1 for $r = 4 \text{ k}\Omega$.

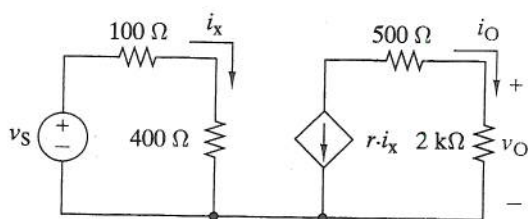


FIGURE P4-1

- 4-2 Find the voltage gain v_O/v_1 and the current gain i_O/i_S in Figure P4-2. For $i_S = 2 \text{ mA}$, find the power supplied by the input current source and the power delivered to the $2\text{-k}\Omega$ load resistor.

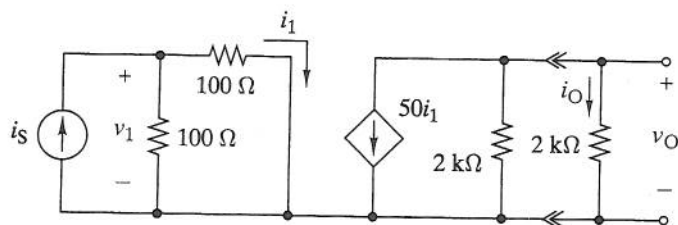


FIGURE P4-2

- 4-3 Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4-3 for $g = 10^{-2} \text{ S}$.

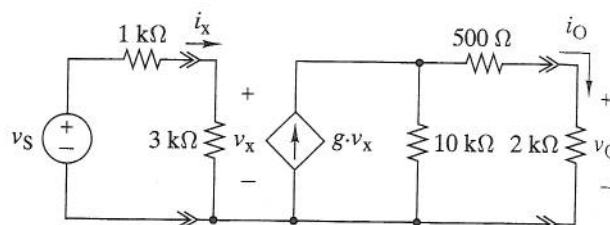


FIGURE P4-3

- 4-4 Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4-4.

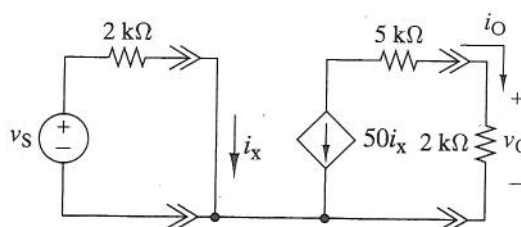


FIGURE P4-4

- 4-5 Find the current gain i_O/i_S in Figure P4-5.

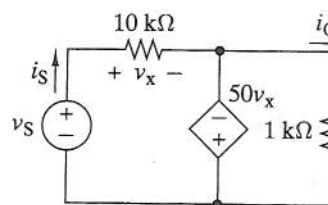


FIGURE P4-5

- 4-6 Find an expression for the current gain i_O/i_S in Figure P4-6. Hint: Apply KCL at node A.

4-7 Find the

10 V

FIG

4-8 Find an
P4-8.
 i_S
 v_S

FIG

4-9 Find an e
P4-9. v_S

F

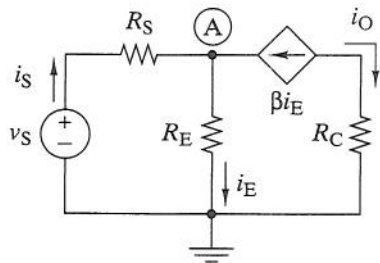


FIGURE P4-6

4-7 Find the voltage v_O in Figure P4-7.

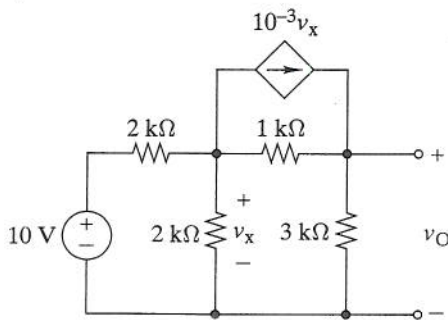


FIGURE P4-7

4-8 Find an expression for the current gain i_O/i_S in Figure P4-8.

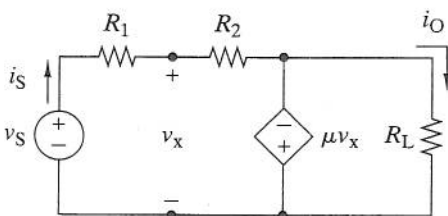


FIGURE P4-8

4-9 Find an expression for the voltage gain v_O/v_S in Figure P4-9.

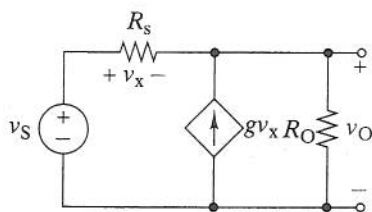


FIGURE P4-9

4-10 Find an expression for the voltage gain v_O/v_S in Figure P4-10.

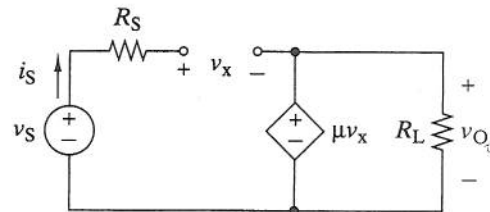


FIGURE P4-10

4-11 Find an expression for the voltage gain v_O/v_S in Figure P4-11.

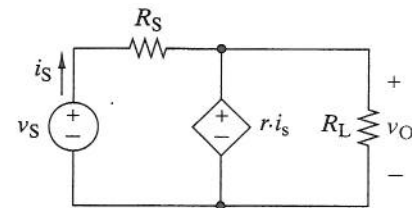


FIGURE P4-11

4-12 Find R_{IN} in Figure P4-12.

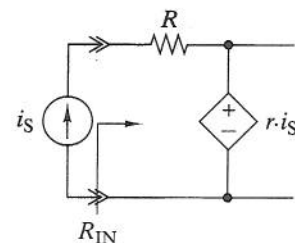


FIGURE P4-12

4-13 Find R_{IN} in Figure P4-13.

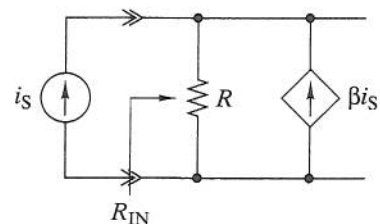


FIGURE P4-13